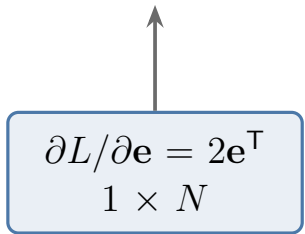
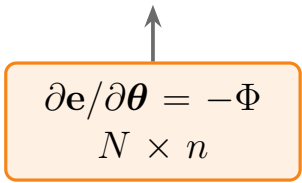
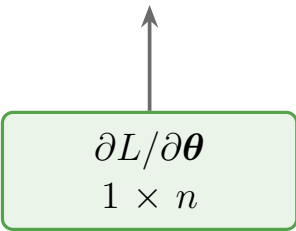


$N = 3$ observations, $n = 2$ parameters: the inner 3s cancel

$$\begin{bmatrix} 2 & 4 & 4 \end{bmatrix} \times \begin{bmatrix} -1 & -1 \\ -1 & -2 \\ -1 & -3 \end{bmatrix} = \begin{bmatrix} -10 & -22 \end{bmatrix}$$


$$\frac{\partial L}{\partial \mathbf{e}} = 2\mathbf{e}^T$$
$$1 \times N$$


$$\frac{\partial \mathbf{e}}{\partial \boldsymbol{\theta}} = -\Phi$$
$$N \times n$$


$$\frac{\partial L}{\partial \boldsymbol{\theta}}$$
$$1 \times n$$

$$\frac{\partial L}{\partial b} = -10,$$
$$\frac{\partial L}{\partial m} = -22$$